BAHRIA UNIVERSITY, (Karachi Campus)

Department of Software Engineering

ASSIGNMENT#02 – Spring 2025

COURSE TITLE:	**Data Encryption and Security**	COURSE CODE:	**CEN-451**
Class:	**BSE – 8A&B**	Shift:	**Morning**
Course Instructor:	**DR. OSAMA REHMAN**	Date:	**12-May-2025**
Due Date:	**19-May-2025**	Marks:	**7.0 Marks**
Name:	**Muhammad Shoaib Akhter Qadri**	Enrollment No:	**02-131212-009**
Section BSE 8B		Reg No:	**79290**

Instructions:

1. **Question#1 & 2 are "Complex Engineering Problems (CEP)" having the attribute of no obvious solution.**
2. **This is an individual effort task.**
3. **Write your full name, registration number and section on the title page.**

Q1. While using Google Chrome as the web-browsing tool, extract the information required as indicated below from the "Digital Certificates" of any two websites of your own choice. Prepare separate tables for each chosen website. **[CLO#5, CEP, 4.0 marks]**

Website: https://blognetwork.live

S#	Parameter	Answer
1	Google Chrome Version	136.0.7103.114 (Official Build) (64-bit)
2	Website URL	https://blognetwork.live
3	Issued to Common Name (CN)	blognetwork.live
4	Issued by Common Name (CN)	E6
5	Issued On	Wednesday 19 March 2025 at 21:03:45
6	Expires On	Tuesday 17 June 2025 at 21:03:44
7	Subject Public Key Algorithm	Elliptic Curve Public Key
8	Subject Public Key Value	00 04 D9 0D 6C F1 41 24 9B BD FA 77 F6 29 0F A3 8B B4 24 25 D9 C8 EF E5 58 07 8B 67 9E 9A 0E 6C A6 BF CA BC 28 86 32 2A 16 E3 13 E1 19 07 97 D5 BE 57 96 8D FC 10 34 CF CB E2 D3 17 47 57 07 A6 32 55 EC DD DD 0A 10 3C 0F 6A BC 35 7B 6C EF 9F D9 3B 44 E3 82 4D B7 DF 3F E5 5B 34 42 B2 66 91 B0 0C
9	Public Key Length *(bits)*	256
10	Public Key Length *(decimal)*	230
11	Hash Function Algorithm	X9.62 ECDSA Signature with SHA-384
12	Device Name / ID *(From Computer)*	VivoBook

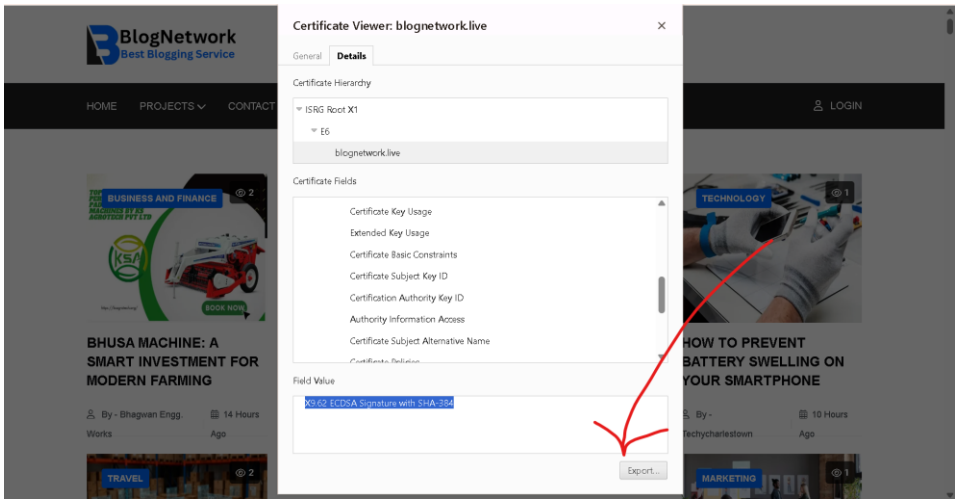
Website: https://universalblog.live

S#	Parameter	Answer
1	Google Chrome Version	136.0.7103.114 (Official Build) (64-bit)
2	Website URL	https://universalblog.live
3	Issued to Common Name (CN)	universalblog.live
4	Issued by Common Name (CN)	R11
5	Issued On	Wednesday 23 April 2025 at 09:04:18
6	Expires On	Tuesday 22 July 2025 at 09:04:17
7	Subject Public Key Algorithm	PKCS #1 RSA Encryption
8	Subject Public Key Value	Modulus (2048 bits): C9 C1 D0 EB 49 33 2F 20 1B 0E 1F 4B AD 3B D4 D2 C6 97 D3 B5 78 DD 59 C4 04 11 93 79 45 B5 5E F8 50 64 27 9D 40 9D 58 85 48 3D E2 A4 FA E9 1B 36 5B DD 0F 27 78 99 24 19 22 49 B2 09 1C 87 8B 65 72 94 9F E0 A3 3D EA C0 3B 8E F4 F6 55 32 D5 20 FC 52 88 59 C7 99 EC AF 68 F3 59 88 FE EE A9 C3 B9 9F 94 AF ED D1 4A 9F 42 B0 DB 24 88 34 2F E6 90 78 C8 40 6D 3E 7B 32 E2 53 E2 B2 95 BE AC 8A 84 6D FC 8D 56 52 E7 38 01 A3 83 7F B4 05 D2 C4 37 FA A7 C2 5E 8C 26 74 83 69 77 D4 46 F9 D1 79 47 F3 61 77 11 40 E1 71 10 9B 04 3C 69 51 57 03 40 BD 9D A6 4B F0 B2 39 7A 90 9F B4 4D 46 72 DE 9C 08 87 2E 2C 3F 21 53 45 8E B9 EA D3 65 FE C5 DC 51 49 E2 47 78 52 54 19 B4 43 FD F8 E1 FE D3 53 A4 74 51 84 6E 41 B7 6E 3F E1 89 55 A9 1B 7F 3C 01 4C 0B 98 3F D0 B3 53 2B 22 87 38 40 E2 05 Public Exponent (17 bits): 01 00 01
9	Public Key Length *(bits)*	2048
10	Public Key Length *(decimal)*	617
11	Hash Function Algorithm	PKCS #1 SHA-256 With RSA Encryption
12	Device Name / ID *(From Computer)*	VivoBook

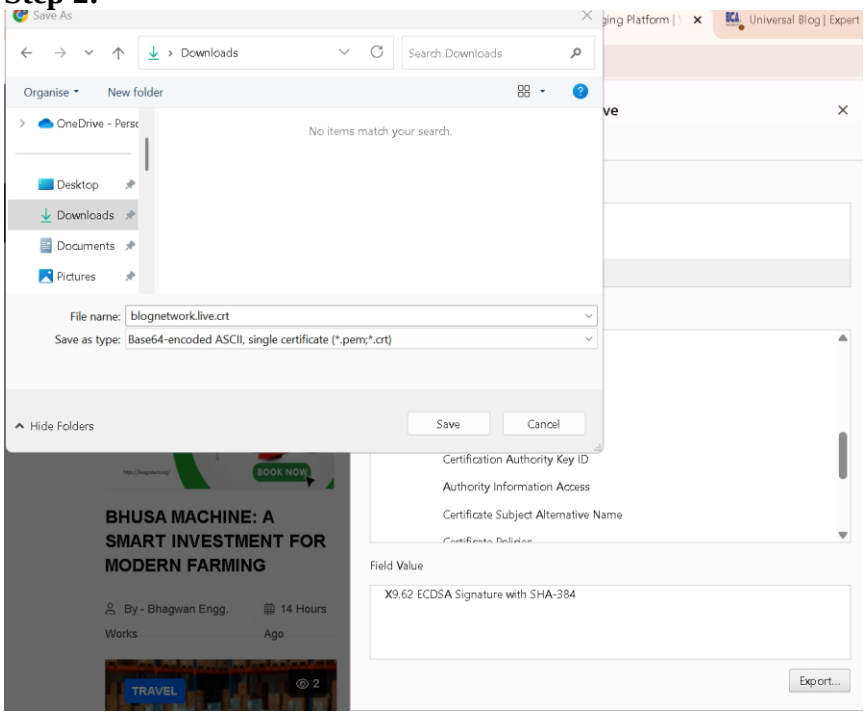
Q2. Choosing one of the websites selected in (Q1), Export its certificate into your device and get it installed into your "current user" account and while using "automatic" selection of location based on certificate type. Once installed, locate where the certificate is installed using Windows "certificate manager". Provide snapshots of the whole process conducted along with snapshot of the location in which the certificate is installed. **[CLO#3, CEP, 3.0 marks]**

Website: https://blognetwork.live:

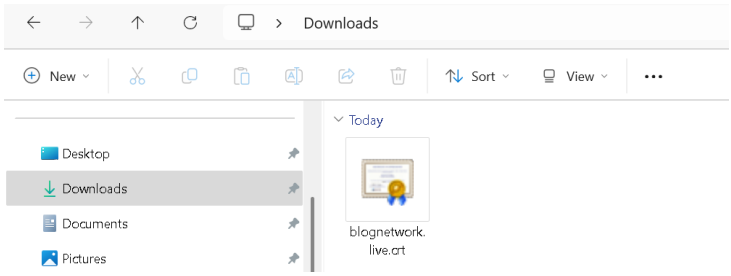
Step 1:


BlogNetwork
Best Blogging Service
HOME
PROJECTS
CONTACT
LOGIN
Certificate Viewer: blognetwork.live
General
Details
Certificate Hierarchy
ISRG Root X1
E6
blognetwork.live
Certificate Fields
Certificate Key Usage
Extended Key Usage
Certificate Basic Constraints
Certificate Subject Key ID
Certification Authority Key ID
Authority Information Access
Certificate Subject Alternative Name
Field Value
X9.62 ECDSA Signature with SHA-384
Export...
BUSINESS AND FINANCE
BHUSA MACHINE: A SMART INVESTMENT FOR MODERN FARMING
TECHNOLOGY
HOW TO PREVENT BATTERY SWELLING ON YOUR SMARTPHONE
TRAVEL
MARKETING

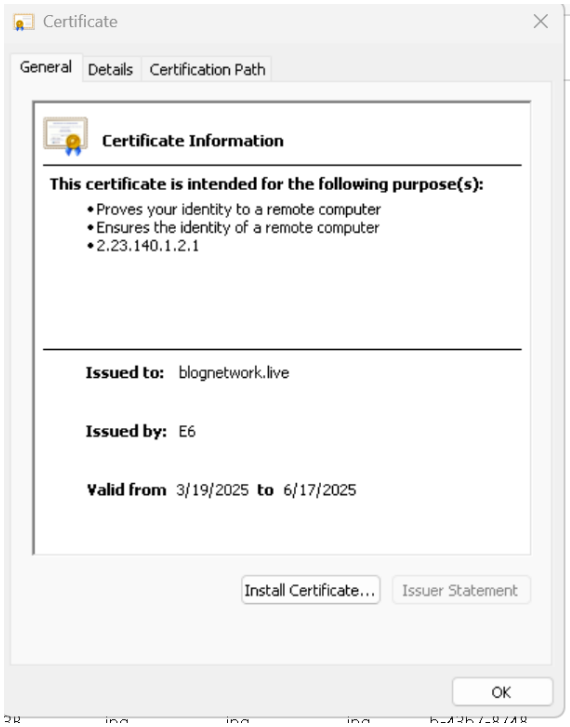
Step 2:


Save As
Downloads
Search Downloads
Organise
New folder
No items match your search.
OneDrive - Perso
Desktop
Downloads
Documents
Pictures
File name: blognetwork.live.crt
Save as type: Base64-encoded ASCII, single certificate (*.pem;*.crt)
Hide Folders
Save
Cancel
Certification Authority Key ID
Authority Information Access
Certificate Subject Alternative Name
Field Value
X9.62 ECDSA Signature with SHA-384
Export...
BHUSA MACHINE: A SMART INVESTMENT FOR MODERN FARMING
TRAVEL

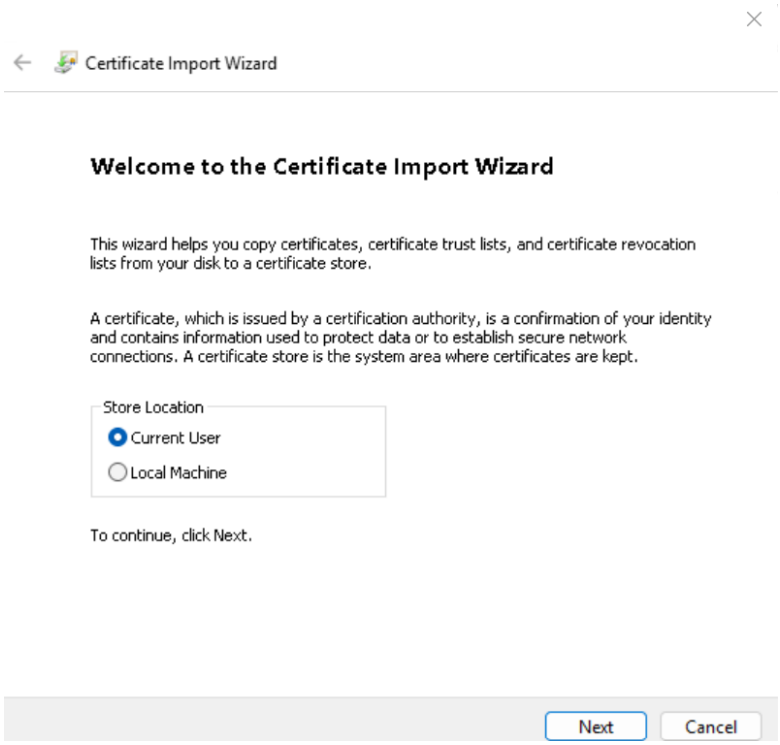
Step 3:


Downloads
New
Sort
View
Today
Desktop
Downloads
Documents
Pictures
blognetwork.live.crt

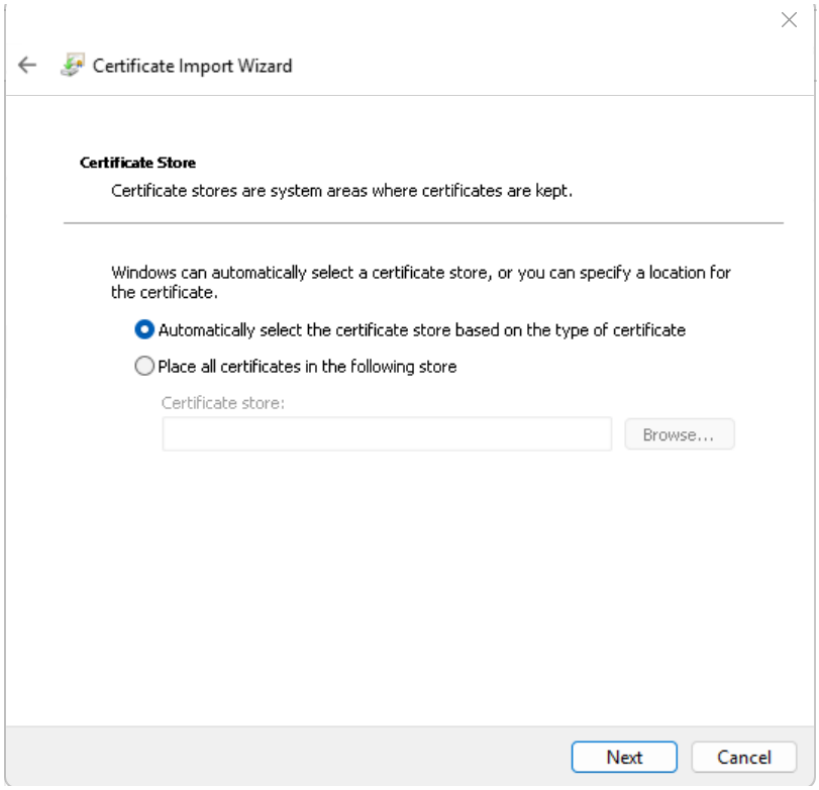
Step 4:



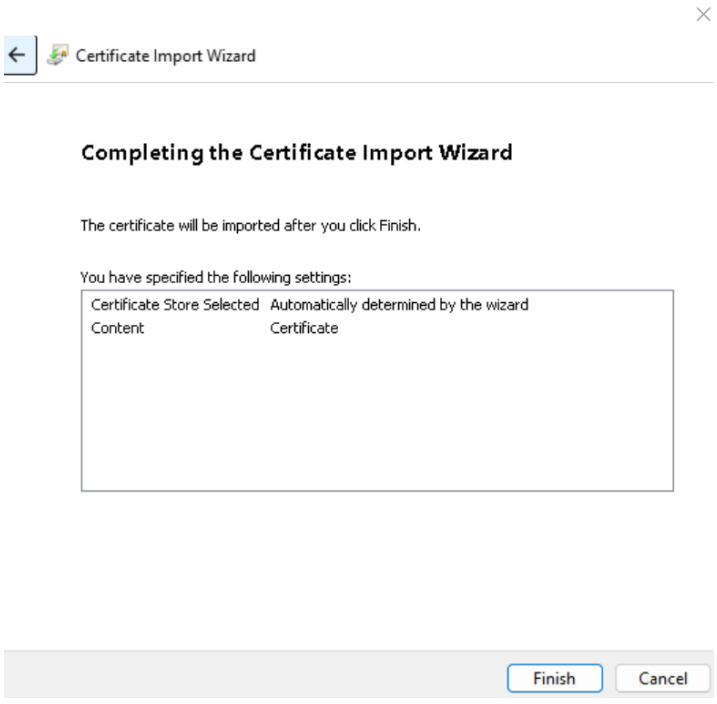
Step 5:



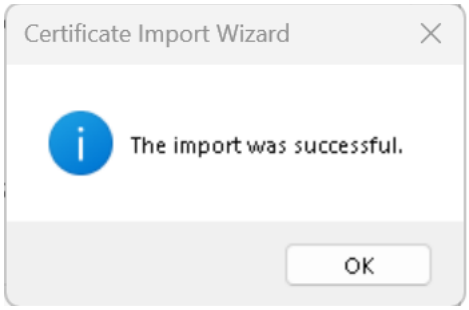
Step 6:


Certificate Import Wizard
Certificate Store
Certificate stores are system areas where certificates are kept.
Windows can automatically select a certificate store, or you can specify a location for the certificate.
Automatically select the certificate store based on the type of certificate
Place all certificates in the following store
Certificate store:
Browse...
Next
Cancel

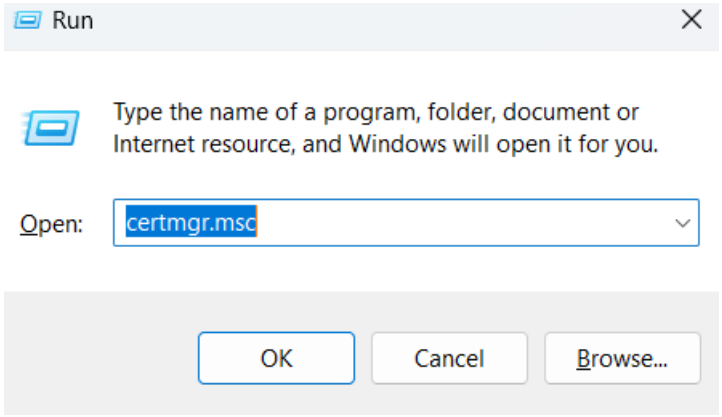
Step 7:


Certificate Import Wizard
Completing the Certificate Import Wizard
The certificate will be imported after you click Finish.
You have specified the following settings:
Certificate Store Selected
Automatically determined by the wizard
Content
Certificate
Finish
Cancel

Step 8:



Step 9:



Step 10:

